

Context Diagram — Peritoneal Dialysis Remote Patient Monitoring System (PD-RPM)

What Is a Context Diagram?

A context diagram is the highest-level model of a system. It represents the system of interest (Sol) as a single undifferentiated *black box*, draws an explicit boundary around it, and shows every external entity the system exchanges something with, together with the labeled, directional flows that cross that boundary. Its defining characteristic is what it deliberately withholds: it says nothing whatsoever about internal components, functions, or architecture. INCOSE defines the system boundary as the “line of demarcation” between the system under consideration and its greater context, and states that the boundary determines what belongs to the system and what does not. The context diagram is the artifact that makes that line explicit, visible, and reviewable by people who would otherwise each assume a different one.

Its notation is intentionally minimal — one box for the Sol, one box per external entity, one labeled arrow per flow. That economy is the point. Because the diagram cannot express internal design, it cannot be used to argue about internal design, which forces a team to finish the question of scope before opening the question of solution.

What Is Its Purpose?

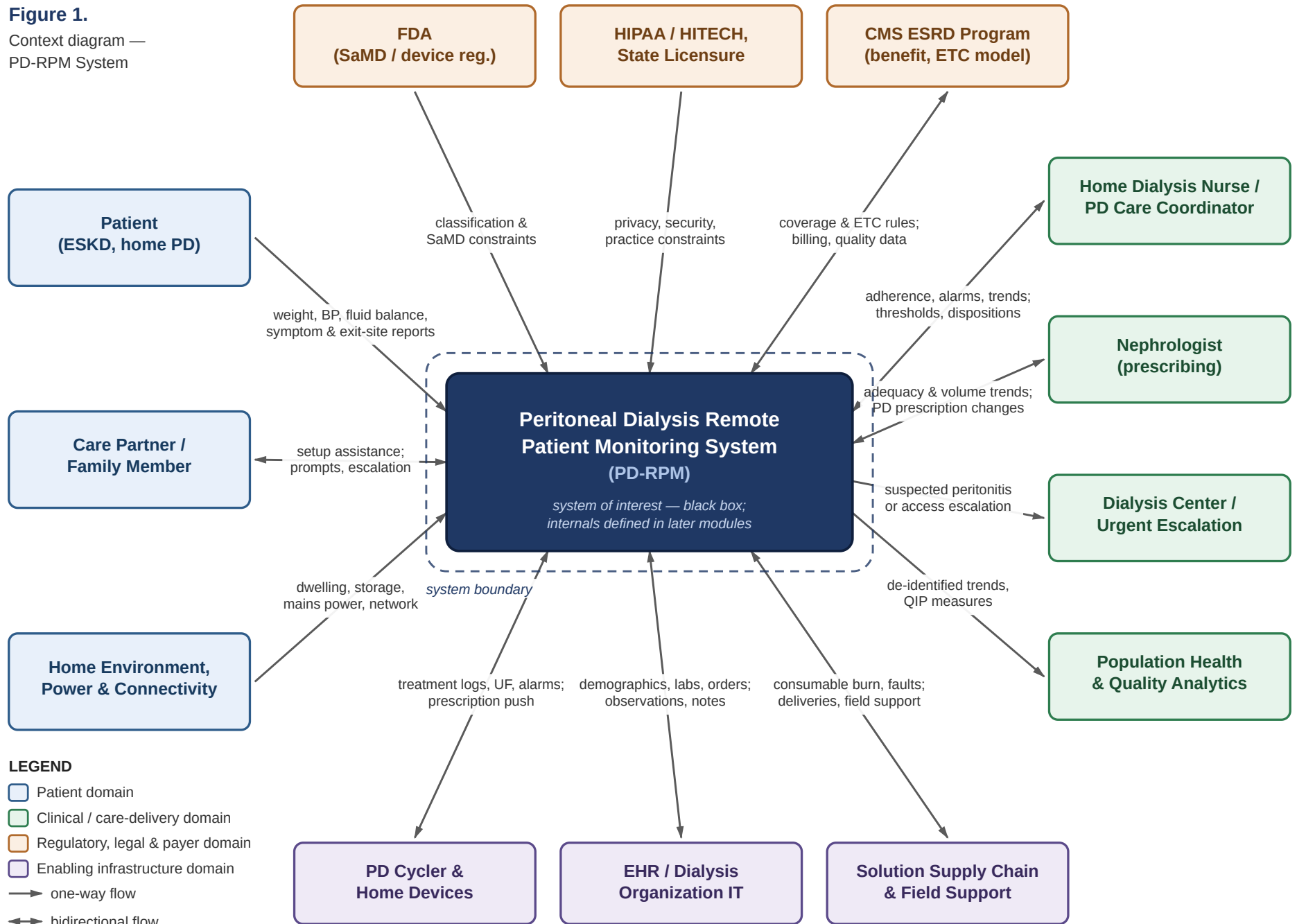
1. **Fix scope and boundary before design begins.** Every element is either inside the box or outside it, and the diagram makes that commitment visible. This is the earliest and cheapest defense against both scope creep and its quieter twin, scope gaps — the interfaces nobody claimed because nobody drew them.
2. **Enumerate the full stakeholder set.** Building the diagram forces the question “who or what else touches this?” until the answers run out. It routinely surfaces entities no single stakeholder would have volunteered — the supply chain, the payer, the care partner — a structural remedy for the blind-men-and-the-elephant problem, in which each participant describes only the part they can see.
3. **Identify every external interface.** Each arrow crossing the boundary is a future interface requirement. Interfaces are where systems most often fail and where integration cost concentrates, so enumerating them at the beginning is disproportionately valuable relative to the effort involved.
4. **Seed operational needs and requirements.** The flows are the raw material for the needs and requirements developed later in the lifecycle: what the system must accept, what it must produce, at what rate, and under what conditions. The diagram is therefore the first input to the systems engineering method.
5. **Expose constraints that are not negotiable.** Laws, regulations, reimbursement rules, and existing infrastructure are not design choices; they are givens that the design must satisfy. Placing them on the diagram prevents a team from discovering them late, when accommodating them is most expensive.
6. **Create a shared mental model.** A single page a clinician, an engineer, a compliance officer, and an executive can all read and correct is how a systems view becomes shared rather than privately held.

Selected Remote Healthcare System

The system of interest is a **peritoneal dialysis remote patient monitoring system (PD-RPM)**: a home-based capability that monitors adults with end-stage kidney disease performing peritoneal dialysis at home and escalates to the home dialysis care team when the data indicate the therapy is failing. National policy has pushed to expand home dialysis — the 2019 Advancing American Kidney Health initiative and CMS’s ESRD Treatment Choices model both incentivize it — yet home PD carries a persistent technique-failure problem, with peritonitis and inadequate ultrafiltration among the leading causes of transfer back to in-center hemodialysis. The signals preceding those failures are generated nightly: declining ultrafiltration volumes, missed or shortened exchanges, recurring cyclor alarms, rising weight and blood pressure, early exit-site changes. They are frequently reviewed only at a monthly clinic visit. The gap is not therapy or diagnostic knowledge; it is a sensing-and-escalation capability operating in the interval between visits. The environment spans this module’s tiers: home-based care, a clinical monitoring node at the dialysis organization, and urgent in-center or emergency response at the escalation limit.

Figure 1.

Context diagram —
PD-RPM System



Every arrow crossing the boundary is a candidate interface requirement.

Interface Summary — Flows Crossing the System Boundary

External Entity	Domain	Dir.	Content of the flow
Patient (ESKD, home PD)	Patient	In	Daily weight, blood pressure, self-reported fluid balance, structured symptom responses, and exit-site condition reports (including images where supported).
Care Partner / Family Member	Patient	Bi	In: assistance with cyclor setup and measurement, confirmation of readings. Out: exchange reminders, plain-language guidance, escalation notification.
Home Environment, Power & Connectivity	Patient	In	Dwelling conditions and solution storage space, mains power (the cyclor runs overnight), and broadband or cellular service of variable reliability. Given conditions, not design choices.
Home Dialysis Nurse / PD Care Coordinator	Clinical	Bi	Out: session adherence, cyclor alarm history, ultrafiltration and volume trends, prioritized triage queue. In: patient-specific thresholds, care-plan updates, alert dispositions.
Nephrologist (prescribing)	Clinical	Bi	Out: adequacy and volume-status trends, escalation summaries, technique-failure risk indicators. In: PD prescription changes (dwell, fill volume, dextrose concentration, cycle count).
Dialysis Center / Urgent Escalation	Clinical	Out	Suspected-peritonitis alert (effluent cloudiness, abdominal pain, fever) or catheter/exit-site access alert, with recent trend context, at the point where remote management is no longer sufficient.
Population Health & Quality Analytics	Clinical	Out	De-identified longitudinal data supporting technique-survival analysis, home-program evaluation, and ESRD Quality Incentive Program measure reporting.
FDA (SaMD / device regulation)	Governance	In	Device classification, software-as-a-medical-device constraints, clinical-decision-support boundaries, labeling and change-control obligations.
HIPAA / HITECH, State Licensure	Governance	In	PHI privacy and security obligations, audit-logging requirements, breach-notification duties, cross-state telehealth practice constraints.
CMS ESRD Program (benefit, ETC model)	Governance	Bi	In: Medicare ESRD benefit coverage rules, ETC model home-dialysis incentives, RPM billing criteria, QIP measure definitions. Out: billing-qualifying data, time documentation, quality measure submissions.
PD Cyclor & Home Devices	Infrastructure	Bi	In: per-session treatment logs (fill and drain volumes, dwell times, net ultrafiltration), alarm records, device state; plus BP cuff and scale readings. Out: prescription updates pushed to the cyclor.
EHR / Dialysis Organization IT	Infrastructure	Bi	In: patient demographics, problem list, medications, laboratory results (Kt/V, albumin, potassium), orders, identity and access control. Out: observations, flowsheet entries, alerts, documentation.
Solution Supply Chain & Field Support	Infrastructure	Bi	Out: consumable burn rate, projected solution shortfall, device fault and service triggers. In: dialysate and supply deliveries, cyclor replacement, on-site or remote technical support.

Dir. — In: entity to system. Out: system to entity. Bi: bidirectional.

Boundary Decisions — What Is Deliberately Outside the Box

A boundary is only meaningful if the exclusions are stated as explicitly as the inclusions. The following are outside the PD-RPM boundary by decision, not by oversight.

Excluded from the Sol	Rationale
The PD cyclor itself	A separately regulated Class II device with its own manufacturer, clearance, and control loop. PD-RPM consumes its telemetry and pushes prescription updates across a defined interface, but does not deliver or modulate therapy.
Home hemodialysis	A distinct modality carrying water treatment, vascular access, and a different alarm set as whole subsystems. Excluded to keep the Sol bounded; a candidate for a later increment rather than this one.
In-center hemodialysis	Not remote care. The center is a destination the system escalates to, not part of the monitored system.
The EHR / dialysis organization clinical information system	PD-RPM exchanges data with it across a defined interface. Absorbing it into the boundary would make the project unbounded.

Clinical decision authority over the PD prescription	The system surfaces trends and escalates; the nephrologist prescribes. This boundary is a regulatory constraint, not a preference.
Solution manufacturing and distribution logistics	PD-RPM signals consumption and faults; the supply organization fulfills. The interface is in scope, the logistics operation is not.

Assumptions

- The deployment setting is home peritoneal dialysis in the United States, for adult ESKD patients enrolled in a dialysis organization's home program.
- Monitoring is continuous for the duration of home PD therapy rather than time-bounded, and ends at modality change, transplant, or death.
- The system is informational and escalatory. It does not deliver therapy and does not exercise autonomous clinical decision authority over the PD prescription.
- Patients and care partners may have limited technical proficiency, intermittent connectivity, and variable support at home. These are context conditions the design must tolerate, not assumptions to be engineered away.
- Early detection of peritonitis and of declining ultrafiltration is taken as the core value hypothesis, since both are leading contributors to technique failure.

References

INCOSE. (2023). *Systems engineering handbook* (5th ed.). INCOSE-TP-2003-002-05. · Kossiakoff, A., Biemer, S. M., Seymour, S. J., & Flanigan, D. A. (2020). *Systems engineering principles and practice* (3rd ed.). Wiley. · Montoya, M. Module 1: Framework, context, and motivation [Course lecture]. Johns Hopkins University. · Executive Order 13879, Advancing American Kidney Health (2019). · Centers for Medicare & Medicaid Services, ESRD Treatment Choices (ETC) Model.